2.42 2nd Order DEs: Homogeneous

Worked Solutions

1 a)

- 2nd order
- Linear
- Homogeneous
- An auxiliary equation can be formed

1 b)

- 2nd order
- Not linear (it contains y^2)
- Homogeneous
- An auxiliary equation cannot be formed

1 c)

- 2nd order
- Linear
- Not homogeneous (RHS $\neq 0$)
- An auxiliary equation cannot be formed (auxiliary equations can only be formed from homogeneous equations)

1 d)

- 3rd order
- Linear
- Homogeneous
- An auxiliary equation can be formed (although it would be a cubic and beyond the scope of this course)

1 e)

This is:

- 2nd order
- Linear
- Homogeneous
- An auxiliary equation cannot be formed (the coefficients are not constant, as x is the coefficient of dy/dx)

1 f)

- 2nd order (note that $\ddot{x} = d^2x/dt^2$ and $\dot{x} = dx/dt$)
- Linear
- Not homogeneous (RHS $\neq 0$)
- An auxiliary equation cannot be formed (auxiliary equations can only be formed from homogeneous equations)

1 g)

- 1st order
- Not linear (contains $(y')^2$)

- Homogeneous
- An auxiliary equation cannot be formed (the equation is not linear)

1 h)

- 2nd order (note that $\ddot{x} = d^2x/dt^2$ and $\dot{x} = dx/dt$)
- Linear
- Homogeneous
- An auxiliary equation can be formed

2 a)

$$
\begin{aligned}
y &= e^{2x} \\
\frac{dy}{dx} &= 2e^{2x} \\
\frac{d^2y}{dx^2} &= 4e^{2x} \\
\frac{d^2y}{dx^2} - 7\frac{dy}{dx} + 10y &= 0 \\
\text{LHS} &= 4e^{2x} - 7(2e^{2x}) + 10e^{2x} \\
&= 4e^{2x} - 14e^{2x} + 10e^{2x} \\
&= (4 - 14 + 10)e^{2x} \\
&= 0 = \text{RHS}.
\end{aligned}
$$

2 b)

$$
\begin{aligned}
y &= 4e^{2x} \\
\frac{dy}{dx} &= 8e^{2x} \\
\frac{d^2y}{dx^2} &= 16e^{2x} \\
\text{LHS} &= 16e^{2x} - 7(8e^{2x}) + 10(4e^{2x}) \\
&= 16e^{2x} - 56e^{2x} + 40e^{2x} \\
&= (16 - 56 + 40)e^{2x} \\
&= 0 = \text{RHS}.
\end{aligned}
$$

2 c)

$$
\begin{aligned}
y &= e^{5x} \\
\frac{dy}{dx} &= 5e^{5x} \\
\frac{d^2y}{dx^2} &= 25e^{5x} \\
\text{LHS} &= 25e^{5x} - 7(5e^{5x}) + 10e^{5x} \\
&= 25e^{2x} - 35e^{2x} + 10e^{2x}
\end{aligned}
$$

$$
\begin{aligned}
&= (25 - 35 + 10)e^{2x} \\
&= 0 = \text{RHS}.
\end{aligned}
$$

2 d)

$$
\begin{aligned}
y &= 7e^{5x} \\
\frac{dy}{dx} &= 35e^{5x} \\
\frac{d^2y}{dx^2} &= 175e^{5x} \\
\text{LHS} &= 175e^{5x} - 7(35e^{5x}) + 10(7e^{5x}) \\
&= 175e^{5x} - 245e^{2x} + 70e^{2x} \\
&= (175 - 245 + 70)e^{2x} \\
&= 0 = \text{RHS}.
\end{aligned}
$$

2 e)

$$
\begin{aligned}
y &= e^{2x} + e^{5x} \\
\frac{dy}{dx} &= 2e^{2x} + 5e^{5x} \\
\frac{d^2y}{dx^2} &= 4e^{2x} + 25e^{5x} \\
\text{LHS} &= (4e^{2x} + 25e^{5x}) - 7(2e^{2x} + 5e^{5x}) + 10(e^{2x} + e^{5x}) \\
&= 4e^{2x} + 25e^{5x} - 14e^{2x} - 35e^{5x} + 10e^{2x} + 10e^{5x} \\
&= (4 - 14 + 10)e^{2x} + (25 - 35 + 10)e^{5x} \\
&= 0 = \text{RHS}.
\end{aligned}
$$

2 f)

$$
\begin{aligned}
y &= 4e^{2x} + 7e^{5x} \\
\frac{dy}{dx} &= 8e^{2x} + 35e^{5x} \\
\frac{d^2y}{dx^2} &= 16e^{2x} + 175e^{5x} \\
\text{LHS} &= (16e^{2x} + 175e^{5x}) - 7(8e^{2x} + 35e^{5x}) + 10(4e^{2x} + 7e^{5x}) \\
&= 16e^{2x} + 175e^{5x} - 56e^{2x} - 245e^{5x} + 40e^{2x} + 70e^{5x} \\
&= (16 - 56 + 40)e^{2x} + (175 - 245 + 70)e^{5x} \\
&= 0 = \text{RHS}.
\end{aligned}
$$

3 a)

$$\begin{aligned} y &= e^{mx} \\ \frac{dy}{dx} &= me^{mx} \\ \frac{d^2y}{dx^2} &= m^2e^{mx} \\ \frac{d^2y}{dx^2} - 7\frac{dy}{dx} + 10y &= 0 \\ m^2e^{mx} - 7me^{mx} + 10e^{mx} &= 0 \\ (m^2 - 7m + 10)e^{mx} &= 0 \\ m^2 - 7m + 10 &= 0, \quad \text{as } e^{mx} \neq 0 \text{ for all } m. \end{aligned}$$

3 b)

$$\begin{aligned} m^2 - 7m + 10 &= 0 \\ (m-2)(m-5) &= 0 \\ m &= 2 \text{ or } m = 5. \end{aligned}$$

Let $m_1 = 2, m_2 = 5$, so $y = Ae^{2x} + Be^{5x}$.

$$\begin{aligned} \frac{dy}{dx} &= 2Ae^{2x} + 5Be^{5x} \\ \frac{d^2y}{dx^2} &= 4Ae^{2x} + 25Be^{5x} \\ \frac{d^2y}{dx^2} - 7\frac{dy}{dx} + 10y &= (4Ae^{2x} + 25Be^{5x}) - 7(2Ae^{2x} + 5Be^{5x}) + 10(Ae^{2x} + Be^{5x}) \\ &= 4Ae^{2x} + 25Be^{5x} - 14Ae^{2x} - 35Be^{5x} + 10Ae^{2x} + 10Be^{5x} \\ &= (4A - 14A + 10A)e^{2x} + (25B - 35B + 10B)e^{5x} \\ &= 0 = \text{RHS}. \end{aligned}$$

4 a)

$$\begin{aligned} \text{Let } y &= e^{mx} \\ \frac{dy}{dx} &= me^{mx} \\ \frac{d^2y}{dx^2} &= m^2e^{mx}. \end{aligned}$$

Then

$$\begin{aligned} \frac{d^2y}{dx^2} - 3\frac{dy}{dx} + 2y &= 0 \\ m^2e^{mx} - 3me^{mx} + 2e^{mx} &= 0 \\ (m^2 - 3m + 2)e^{mx} &= 0. \end{aligned}$$

As $e^{mx} \neq 0$, the auxiliary equation is

$$\begin{aligned} m^2 - 3m + 2 &= 0 \\ (m-1)(m-2) &= 0 \\ m = 1 \text{ or } m &= 2. \end{aligned}$$

General solution: $y = Ae^x + Be^{2x}$.

4 b)

$$\begin{aligned} \text{Let } y &= e^{mx} \\ \frac{dy}{dx} &= me^{mx} \\ \frac{d^2y}{dx^2} &= m^2e^{mx}. \end{aligned}$$

Then

$$\begin{aligned} \frac{d^2y}{dx^2} - 2\frac{dy}{dx} - 35y &= 0 \\ m^2e^{mx} - 2me^{mx} - 35e^{mx} &= 0 \\ (m^2 - 2m - 35)e^{mx} &= 0. \end{aligned}$$

As $e^{mx} \neq 0$, auxiliary equation:

$$\begin{aligned} m^2 - 2m - 35 &= 0 \\ (m-7)(m+5) &= 0 \\ m = 7 \text{ or } m &= -5. \end{aligned}$$

General solution: $y = Ae^{7x} + Be^{-5x}$.

4 c)

$$\begin{aligned} \text{Let } y &= e^{mx} \\ \frac{dy}{dx} &= me^{mx} \\ \frac{d^2y}{dx^2} &= m^2e^{mx}. \end{aligned}$$

Then

$$\begin{aligned} \frac{d^2y}{dx^2} + 18\frac{dy}{dx} + 80y &= 0 \\ m^2e^{mx} + 18me^{mx} + 80e^{mx} &= 0 \\ (m^2 + 18m + 80)e^{mx} &= 0. \end{aligned}$$

As $e^{mx} \neq 0$, auxiliary equation:

$$\begin{aligned} m^2 + 18m + 80 &= 0 \\ (m+10)(m+8) &= 0 \\ m = -10 \text{ or } m &= -8. \end{aligned}$$

General solution: $y = Ae^{-10x} + Be^{-8x}$.

4 d)

$$\begin{aligned}\text{Let } y &= e^{mx} \\ \frac{dy}{dx} &= me^{mx} \\ \frac{d^2y}{dx^2} &= m^2e^{mx}.\end{aligned}$$

Then

$$\begin{aligned}2\frac{d^2y}{dx^2} + 11\frac{dy}{dx} - 6y &= 0 \\ 2m^2e^{mx} + 11me^{mx} - 6e^{mx} &= 0 \\ (2m^2 + 11m - 6)e^{mx} &= 0.\end{aligned}$$

As $e^{mx} \neq 0$, auxiliary equation:

$$\begin{aligned}2m^2 + 11m - 6 &= 0 \\ (2m-1)(m+6) &= 0 \\ m = \frac{1}{2} &\text{ or } m = -6.\end{aligned}$$

General solution: $y = Ae^{x/2} + Be^{-6x}$.

4 e)

$$\begin{aligned}\text{Let } y &= e^{mx} \\ \frac{dy}{dx} &= me^{mx} \\ \frac{d^2y}{dx^2} &= m^2e^{mx}.\end{aligned}$$

Then

$$\begin{aligned}\frac{d^2y}{dx^2} - 4y &= 0 \\ m^2e^{mx} - 4e^{mx} &= 0 \\ (m^2 - 4)e^{mx} &= 0.\end{aligned}$$

As $e^{mx} \neq 0$, auxiliary equation:

$$\begin{aligned}m^2 - 4 &= 0 \\ (m-2)(m+2) &= 0 \\ m = 2 &\text{ or } m = -2.\end{aligned}$$

General solution: $y = Ae^{2x} + Be^{-2x}$.

4 f)

$$\text{Let } y = e^{mx}$$

$$\frac{dy}{dx} = me^{mx}$$
$$\frac{d^2y}{dx^2} = m^2e^{mx}.$$

Then

$$\frac{d^2y}{dx^2} - 4\frac{dy}{dx} = 0$$
$$m^2e^{mx} - 4me^{mx} = 0$$
$$(m^2 - 4m)e^{mx} = 0.$$

As $e^{mx} \neq 0$, auxiliary equation:

$$m^2 - 4m = 0$$
$$(m-4)m = 0$$
$$m = 4 \text{ or } m = 0.$$

General solution:

$$y = Ae^{4x} + Be^0$$
$$= Ae^{4x} + B.$$

5 a)

$$\ddot{x} + 3\dot{x} - 28x = 0$$
$$m^2 + 3m - 28 = 0$$
$$(m+7)(m-4) = 0$$
$$m = -7 \text{ or } m = 4$$
$$x = Ae^{-7t} + Be^{4t}.$$

5 b)

$$4\ddot{x} + 4\dot{x} - 15x = 0$$
$$4m^2 + 4m - 15 = 0$$
$$(2m+5)(2m-3) = 0$$
$$m = -\frac{5}{2} \text{ or } m = \frac{3}{2}$$
$$x = Ae^{-5t/2} + Be^{3t/2}.$$

6)

$$\frac{d^2y}{dx^2} + \frac{dy}{dx} - 2y = 0$$
$$m^2 + m - 2 = 0$$
$$(m+2)(m-1) = 0$$

$$m = -2 \text{ or } m = 1$$
$$y = Ae^{-2x} + Be^{x}.$$

Substitute $y = 2, x = 0$:

$$2 = Ae^{0} + Be^{0}$$
$$A + B = 2. \tag{1}$$

Substitute $y = 3, x = \ln 2$:

$$3 = Ae^{-2\ln 2} + Be^{\ln 2}$$
$$= Ae^{\ln 2^{-2}} + Be^{\ln 2}$$
$$\frac{1}{4}A + 2B = 3. \tag{2}$$

Solving (1) and (2), $A = \frac{4}{7}$ and $B = \frac{10}{7}$. Particular solution:

$$y = \frac{4}{7}e^{-2x} + \frac{10}{7}e^{x}.$$

7)

$$2y'' - 7y' + 3y = 0$$
$$2m^2 - 7m + 3 = 0$$
$$(2m - 1)(m - 3) = 0$$
$$m = \frac{1}{2} \text{ or } m = 3$$
$$y = Ae^{x/2} + Be^{3x}.$$

Substitute $y = 5, x = 0$:

$$5 = Ae^{0} + Be^{0}$$
$$A + B = 5. \tag{1}$$

Substitute $y = -238, x = \ln 4$:

$$-238 = Ae^{\frac{1}{2}\ln 4} + Be^{3\ln 4}$$
$$= Ae^{\ln 4^{1/2}} + Be^{\ln 4^{3}}$$
$$2A + 64B = -238. \tag{2}$$

Solving (1) and (2), $A = 9$ and $B = -4$. Particular solution:

$$y = 9e^{x/2} - 4e^{3x}.$$

8 a)

$$y = e^{3x}$$
$$\frac{dy}{dx} = 3e^{3x}$$

$$\frac{d^2y}{dx^2} = 9e^{3x}$$

$$\frac{d^2y}{dx^2} - 6\frac{dy}{dx} + 9y = 0$$

$$\begin{aligned}\text{LHS} &= 9e^{3x} - 6(3e^{3x}) + 9e^{3x} \\ &= 9e^{3x} - 18e^{3x} + 9e^{3x} \\ &= (9 - 18 + 9)e^{3x} \\ &= 0 = \text{RHS}.\end{aligned}$$

8 b)

$$\begin{aligned}y &= 2e^{3x} \\ \frac{dy}{dx} &= 6e^{3x} \\ \frac{d^2y}{dx^2} &= 18e^{3x} \\ \text{LHS} &= 18e^{3x} - 6(6e^{3x}) + 9(2e^{3x}) \\ &= 18e^{3x} - 36e^{3x} + 18e^{3x} \\ &= (18 - 36 + 18)e^{3x} \\ &= 0 = \text{RHS}.\end{aligned}$$

8 c)

$$\begin{aligned}y &= xe^{3x} \\ \frac{dy}{dx} &= 3xe^{3x} + e^{3x} \\ \frac{d^2y}{dx^2} &= 9xe^{3x} + 3e^{3x} + 3e^{3x} \\ &= 9xe^{3x} + 6e^{3x} \\ \text{LHS} &= (9xe^{3x} + 6e^{3x}) - 6(3xe^{3x} + e^{3x}) + 9xe^{3x} \\ &= 9xe^{3x} + 6e^{3x} - 18xe^{3x} - 6e^{3x} + 9xe^{3x} \\ &= (9 - 18 + 9)xe^{3x} + (6 - 6)e^{3x} \\ &= 0 = \text{RHS}.\end{aligned}$$

8 d)

$$\begin{aligned}y &= 5xe^{3x} \\ \frac{dy}{dx} &= 15xe^{3x} + 5e^{3x} \\ \frac{d^2y}{dx^2} &= 45xe^{3x} + 15e^{3x} + 15e^{3x} \\ &= 45xe^{3x} + 30e^{3x} \\ \text{LHS} &= (45xe^{3x} + 30e^{3x}) - 6(15xe^{3x} + 5e^{3x}) + 9(5xe^{3x})\end{aligned}$$

$$
\begin{aligned}
&= 45xe^{3x} + 30e^{3x} - 90xe^{3x} - 30e^{3x} + 45xe^{3x} \\
&= (45 - 90 + 45)xe^{3x} + (30 - 30)e^{3x} \\
&= 0 = \text{RHS}.
\end{aligned}
$$

8 e)

$$
\begin{aligned}
y &= e^{3x} + xe^{3x} \\
\frac{dy}{dx} &= 3e^{3x} + 3xe^{3x} + e^{3x} \\
&= 3xe^{3x} + 4e^{3x} \\
\frac{d^2y}{dx^2} &= 9xe^{3x} + 3e^{3x} + 12e^{3x} \\
&= 9xe^{3x} + 15e^{3x} \\
\text{LHS} &= (9xe^{3x} + 15e^{3x}) - 6(3xe^{3x} + 4e^{3x}) + 9(e^{3x} + xe^{3x}) \\
&= 9xe^{3x} + 15e^{3x} - 18xe^{3x} - 24e^{3x} + 9e^{3x} + 9xe^{3x} \\
&= (9 - 18 + 9)xe^{3x} + (15 - 24 + 9)e^{3x} \\
&= 0 = \text{RHS}.
\end{aligned}
$$

8 f)

$$
\begin{aligned}
y &= 2e^{3x} + 5xe^{3x} \\
\frac{dy}{dx} &= 6e^{3x} + 15xe^{3x} + 5e^{3x} \\
&= 15xe^{3x} + 11e^{3x} \\
\frac{d^2y}{dx^2} &= 45xe^{3x} + 15e^{3x} + 33e^{3x} \\
&= 45xe^{3x} + 48e^{3x} \\
\text{LHS} &= (45xe^{3x} + 48e^{3x}) - 6(15xe^{3x} + 11e^{3x}) + 9(2e^{3x} + 5xe^{3x}) \\
&= 45xe^{3x} + 48e^{3x} - 90xe^{3x} - 66e^{3x} + 18e^{3x} + 45xe^{3x} \\
&= (45 - 90 + 45)xe^{3x} + (48 - 66 + 18)e^{3x} \\
&= 0 = \text{RHS}.
\end{aligned}
$$

9 a)

$$
\begin{aligned}
y &= xe^{mx} \\
\frac{dy}{dx} &= mxe^{mx} + e^{mx} \\
\frac{d^2y}{dx^2} &= m^2xe^{mx} + me^{mx} + me^{mx} \\
\frac{d^2y}{dx^2} - 6\frac{dy}{dx} + 9y &= 0 \\
(m^2xe^{mx} + me^{mx} + me^{mx}) - 6(mxe^{mx} + e^{mx}) + 9(xe^{mx}) &= 0
\end{aligned}
$$

$$m^2xe^{mx} + me^{mx} + me^{mx} - 6mxe^{mx} - 6e^{mx} + 9xe^{mx} = 0$$
$$(m^2 - 6m + 9)xe^{mx} + (2m - 6)e^{mx} = 0$$
$$((m^2 - 6m + 9)x + (2m - 6))e^{mx} = 0$$
$$m^2 - 6m + 9 = 0 \quad \text{and} \quad 2m - 6 = 0, \text{ as } e^{mx} \neq 0.$$

9 b)

Firstly,

$$m^2 - 6m + 9 = 0 \Rightarrow (m-3)^2 = 0 \Rightarrow m = 3.$$

Secondly, $2m - 6 = 0 \Rightarrow m = 3$, so $m = 3$ satisfies both equations. Let $y = Axe^{3x} + Be^{3x}$.

$$\frac{dy}{dx} = 3Axe^{3x} + Ae^{3x} + 3Be^{3x}$$
$$\frac{d^2y}{dx^2} = 9Axe^{3x} + 3Ae^{3x} + 3Ae^{3x} + 9Be^{3x}$$
$$\begin{aligned}\text{LHS} &= (9Axe^{3x} + 3Ae^{3x} + 3Ae^{3x} + 9Be^{3x}) \\ &\quad - 6(3Axe^{3x} + Ae^{3x} + 3Be^{3x}) + 9(Axe^{3x} + Be^{3x}) \\ &= 9Axe^{3x} + 3Ae^{3x} + 3Ae^{3x} + 9Be^{3x} - 18Axe^{3x} - 6Ae^{3x} - 18Be^{3x} \\ &\quad + 9Axe^{3x} + 9Be^{3x} \\ &= (9A - 18A + 9A)xe^{3x} + (3A + 3A + 9B - 6A - 18B + 9B)e^{3x} \\ &= 0 = \text{RHS}.\end{aligned}$$

10 a)

$$y = e^{mx}, \quad \frac{dy}{dx} = me^{mx}, \quad \frac{d^2y}{dx^2} = m^2e^{mx}$$
$$\text{Then } \frac{d^2y}{dx^2} - 10\frac{dy}{dx} + 25y = 0$$
$$m^2e^{mx} - 10me^{mx} + 25e^{mx} = 0$$
$$(m^2 - 10m + 25)e^{mx} = 0, \quad e^{mx} \neq 0$$
$$m^2 - 10m + 25 = 0$$
$$(m-5)^2 = 0, \quad m = 5$$
$$y = Axe^{5x} + Be^{5x}.$$

10 b)

$$y = e^{mx}, \quad \frac{dy}{dx} = me^{mx}, \quad \frac{d^2y}{dx^2} = m^2e^{mx}$$
$$\text{Then } \frac{d^2y}{dx^2} + 12\frac{dy}{dx} + 36y = 0$$
$$m^2e^{mx} + 12me^{mx} + 36e^{mx} = 0$$
$$(m^2 + 12m + 36)e^{mx} = 0, \quad e^{mx} \neq 0$$
$$m^2 + 12m + 36 = 0$$
$$(m+6)^2 = 0, \quad m = -6$$
$$y = Axe^{-6x} + Be^{-6x}.$$

10 c)

$$y = e^{mx}, \quad \frac{dy}{dx} = me^{mx}, \quad \frac{d^2y}{dx^2} = m^2e^{mx}$$

$$\text{Then } 4\frac{d^2y}{dx^2} + 12\frac{dy}{dx} + 9y = 0$$
$$4m^2e^{mx} + 12me^{mx} + 9e^{mx} = 0$$
$$(4m^2 + 12m + 9)e^{mx} = 0, \quad e^{mx} \neq 0$$
$$4m^2 + 12m + 9 = 0$$
$$(2m + 3)^2 = 0, \quad m = -\frac{3}{2}$$
$$y = Axe^{-3x/2} + Be^{-3x/2}.$$

10 d)

$$y = e^{mx}, \quad \frac{dy}{dx} = me^{mx}, \quad \frac{d^2y}{dx^2} = m^2e^{mx}$$

$$\text{Then } 9\frac{d^2y}{dx^2} - 42\frac{dy}{dx} + 49y = 0$$
$$9m^2e^{mx} - 42me^{mx} + 49e^{mx} = 0$$
$$(9m^2 - 42m + 49)e^{mx} = 0, \quad e^{mx} \neq 0$$
$$9m^2 - 42m + 49 = 0$$
$$(3m - 7)^2 = 0, \quad m = \frac{7}{3}$$
$$y = Axe^{7x/3} + Be^{7x/3}.$$

11 a)

$$y'' + 26y' + 169y = 0$$
$$m^2 + 26m + 169 = 0$$
$$(m + 13)^2 = 0, \quad m = -13$$
$$y = Axe^{-13x} + Be^{-13x}.$$

11 b)

$$9y'' - 48y' + 64y = 0$$
$$9m^2 - 48m + 64 = 0$$
$$(3m - 8)^2 = 0, \quad m = \frac{8}{3}$$
$$y = Axe^{8x/3} + Be^{8x/3}.$$

12)

$$\ddot{x} - 10\dot{x} + 25x = 0$$
$$m^2 - 10m + 25 = 0$$
$$(m-5)^2 = 0, \quad m = 5$$
$$x = Ate^{5t} + Be^{5t}.$$

Substitute $x = 0, t = 0$:

$$0 = A(0)e^0 + Be^0 \Rightarrow B = 0.$$

Substitute $x = 4, t = 1$:

$$4 = A(1)e^{5(1)} + 0 \Rightarrow A = 4e^{-5}.$$

Particular solution:

$$y = 4e^{-5}te^{5t} = 4te^{5t-5}.$$

13)

$$9\frac{d^2y}{dx^2} - 6\frac{dy}{dx} + y = 0$$
$$9m^2 - 6m + 1 = 0$$
$$(3m-1)^2 = 0, \quad m = \frac{1}{3}$$
$$y = Axe^{x/3} + Be^{x/3}.$$

Substitute $y = 3, x = 3$:

$$3 = A(3)e^{(1/3)(3)} + Be^{(1/3)(3)}$$
$$3A + B = \frac{3}{e}. \tag{1}$$

Substitute $y = 4, x = 6$:

$$4 = A(6)e^{(1/3)(6)} + Be^{(1/3)(6)}$$
$$6A + B = \frac{4}{e^2}. \tag{2}$$

Subtract (1) from (2):

$$3A = \frac{4}{e^2} - \frac{3}{e}$$
$$A = \frac{4}{3e^2} - \frac{1}{e} = \frac{4-3e}{3e^2}.$$

Substitute into (1):

$$3\left(\frac{4-3e}{3e^2}\right) + B = \frac{3}{e}$$
$$\frac{4-3e}{e^2} + B = \frac{3}{e}$$
$$B = \frac{3}{e} - \frac{4-3e}{e^2} = \frac{3e-(4-3e)}{e^2} = \frac{6e-4}{e^2}.$$

Particular solution:

$$y = \left(\frac{4-3e}{3e^2}\right)xe^{x/3} + \left(\frac{6e-4}{e^2}\right)e^{x/3}$$
$$= \frac{1}{3}(4-3e)xe^{x/3-2} + (6e-4)e^{x/3-2}.$$

14 a)

$$\begin{aligned} y &= e^{2ix} \\ \frac{dy}{dx} &= 2ie^{2ix} \\ \frac{d^2y}{dx^2} &= -4e^{2ix} \\ \text{LHS} &= -4e^{2ix} + 4e^{2ix} \\ &= 0 = \text{RHS}. \end{aligned}$$

14 b)

$$\begin{aligned} y &= e^{-2ix} \\ \frac{dy}{dx} &= -2ie^{-2ix} \\ \frac{d^2y}{dx^2} &= -4e^{-2ix} \\ \text{LHS} &= -4e^{-2ix} + 4e^{-2ix} \\ &= 0 = \text{RHS}. \end{aligned}$$

14 c)

$$\begin{aligned} y &= 2e^{2ix} - 3e^{-2ix} \\ \frac{dy}{dx} &= 4ie^{2ix} + 6ie^{-2ix} \\ \frac{d^2y}{dx^2} &= -8e^{2ix} + 12e^{-2ix} \\ \text{LHS} &= (-8e^{2ix} + 12e^{-2ix}) + 4(2e^{2ix} - 3e^{-2ix}) \\ &= -8e^{2ix} + 12e^{-2ix} + 8e^{2ix} - 12e^{-2ix} \\ &= 0 = \text{RHS}. \end{aligned}$$

15)

$$\begin{aligned} e^{2ix} &= \cos 2x + i \sin 2x \\ e^{-2ix} &= \cos(-2x) + i \sin(-2x) \\ &= \cos 2x - i \sin 2x \\ Ae^{2ix} + Be^{-2ix} &= A(\cos 2x + i \sin 2x) + B(\cos 2x - i \sin 2x) \\ &= (A + B) \cos 2x + (A - B)i \sin 2x. \end{aligned}$$

Since the differential equation has real coefficients, for real-valued solutions choose A and B as complex conjugates. Let $A = p + qi$ and $B = p - qi$. Then

$$C = A + B = p + qi + p - qi = 2p,$$

$$D = (A - B)i = (p + qi - p + qi)i = -2q.$$

So C and D are real constants, and

$$Ae^{2ix} + Be^{-2ix} = C\cos 2x + D\sin 2x.$$

16 a)

$$\begin{aligned} y &= \cos 2x \\ \frac{dy}{dx} &= -2\sin 2x \\ \frac{d^2y}{dx^2} &= -4\cos 2x \\ \text{LHS} &= -4\cos 2x + 4\cos 2x \\ &= 0 = \text{RHS}. \end{aligned}$$

16 b)

$$\begin{aligned} y &= \sin 2x \\ \frac{dy}{dx} &= 2\cos 2x \\ \frac{d^2y}{dx^2} &= -4\sin 2x \\ \text{LHS} &= -4\sin 2x + 4\sin 2x \\ &= 0 = \text{RHS}. \end{aligned}$$

16 c)

$$\begin{aligned} y &= 3\cos 2x \\ \frac{dy}{dx} &= -6\sin 2x \\ \frac{d^2y}{dx^2} &= -12\cos 2x \\ \text{LHS} &= -12\cos 2x + 4(3\cos 2x) \\ &= 0 = \text{RHS}. \end{aligned}$$

16 d)

$$\begin{aligned} y &= 5\sin 2x \\ \frac{dy}{dx} &= 10\cos 2x \\ \frac{d^2y}{dx^2} &= -20\sin 2x \\ \text{LHS} &= -20\sin 2x + 4(5\sin 2x) \\ &= 0 = \text{RHS}. \end{aligned}$$

16 e)

$$\begin{aligned} y &= \cos 2x + \sin 2x \\ \frac{dy}{dx} &= -2\sin 2x + 2\cos 2x \\ \frac{d^2y}{dx^2} &= -4\cos 2x - 4\sin 2x \\ \text{LHS} &= (-4\cos 2x - 4\sin 2x) + 4(\cos 2x + \sin 2x) \\ &= -4\cos 2x - 4\sin 2x + 4\cos 2x + 4\sin 2x \\ &= 0 = \text{RHS}. \end{aligned}$$

16 f)

$$\begin{aligned} y &= 3\cos 2x + 5\sin 2x \\ \frac{dy}{dx} &= -6\sin 2x + 10\cos 2x \\ \frac{d^2y}{dx^2} &= -12\cos 2x - 20\sin 2x \\ \text{LHS} &= (-12\cos 2x - 20\sin 2x) + 4(3\cos 2x + 5\sin 2x) \\ &= -12\cos 2x - 20\sin 2x + 12\cos 2x + 20\sin 2x \\ &= 0 = \text{RHS}. \end{aligned}$$

17 a)

$$\begin{aligned} y &= e^{mx} \\ \frac{dy}{dx} &= me^{mx} \\ \frac{d^2y}{dx^2} &= m^2e^{mx} \\ \frac{d^2y}{dx^2} + 9y &= 0 \\ m^2e^{mx} + 9e^{mx} &= 0 \\ (m^2+9)e^{mx} &= 0 \\ m^2 + 9 &= 0, \quad \text{as } e^{mx} \neq 0 \text{ for all } m. \end{aligned}$$

17 b)

$$\begin{aligned} m^2 + 9 = 0 \Rightarrow m &= \pm 3i, \\ y &= A\cos 3x + B\sin 3x, \\ \frac{dy}{dx} &= -3A\sin 3x + 3B\cos 3x, \\ \frac{d^2y}{dx^2} &= -9A\cos 3x - 9B\sin 3x, \end{aligned}$$

$$\begin{aligned}\text{LHS} &= (-9A\cos 3x - 9B\sin 3x) + 9(A\cos 3x + B\sin 3x)\\ &= -9A\cos 3x - 9B\sin 3x + 9A\cos 3x + 9B\sin 3x\\ &= 0 = \text{RHS}.\end{aligned}$$

18 a)

$$\begin{aligned}y = e^{mx}, \quad y' = me^{mx}, \quad y'' = m^2e^{mx}\\ y'' + 25y = 0 \Rightarrow (m^2+25)e^{mx} = 0\\ m^2 + 25 = 0, \quad m = \pm 5i\\ y = A\cos 5x + B\sin 5x.\end{aligned}$$

18 b)

$$\begin{aligned}y = e^{mx}, \quad y' = me^{mx}, \quad y'' = m^2e^{mx}\\ y'' + 49y = 0 \Rightarrow (m^2+49)e^{mx} = 0\\ m^2 + 49 = 0, \quad m = \pm 7i\\ y = A\cos 7x + B\sin 7x.\end{aligned}$$

18 c)

$$\begin{aligned}y = e^{mx}, \quad y' = me^{mx}, \quad y'' = m^2e^{mx}\\ 4y'' + 9y = 0 \Rightarrow (4m^2+9)e^{mx} = 0\\ 4m^2 + 9 = 0, \quad m = \pm\frac{3}{2}i\\ y = A\cos\frac{3}{2}x + B\sin\frac{3}{2}x.\end{aligned}$$

19 a)

$$\begin{aligned}&\ddot{x} + 8x = 0\\ &m^2 + 8 = 0, \quad m = \pm 2\sqrt{2}i\\ &\quad x = A\cos(2\sqrt{2}t) + B\sin(2\sqrt{2}t).\end{aligned}$$

19 b)

$$\begin{aligned}&9\ddot{x} + 25x = 0\\ &9m^2 + 25 = 0, \quad m = \pm\frac{5}{3}i\\ &\quad x = A\cos\frac{5}{3}t + B\sin\frac{5}{3}t.\end{aligned}$$

20)

$$\begin{aligned} y'' + 100y &= 0 \\ m^2 + 100 = 0, \quad m &= \pm 10i \\ y &= A\cos 10x + B\sin 10x. \end{aligned}$$

Substitute $y = 0, x = 0$: $0 = A$, so $y = B\sin 10x$. Substitute $y = 3, x = \pi/20$:

$$3 = B\sin(10\pi/20) = B\sin(\pi/2) \Rightarrow B = 3.$$

Particular solution $y = 3\sin 10x$.

21)

$$\begin{aligned} 16y'' + y &= 0 \\ 16m^2 + 1 = 0, \quad m &= \pm\frac{1}{4}i \\ y &= A\cos\frac{x}{4} + B\sin\frac{x}{4}. \end{aligned}$$

Substitute $y = 2, x = \pi$:

$$2 = A\cos\frac{\pi}{4} + B\sin\frac{\pi}{4} = \frac{A+B}{\sqrt{2}}, \quad A + B = 2\sqrt{2}. \tag{1}$$

Substitute $y = -3, x = 2\pi$:

$$-3 = A\cos\frac{\pi}{2} + B\sin\frac{\pi}{2} = B, \quad B = -3. \tag{2}$$

From (1) and (2), $A = 2\sqrt{2} + 3$. Thus

$$y = (2\sqrt{2} + 3)\cos\frac{x}{4} - 3\sin\frac{x}{4}.$$

22 a)

$$\begin{aligned} y &= e^{mx} \\ y' &= me^{mx} \\ y'' &= m^2e^{mx} \\ y'' - 4y' + 13y &= 0 \\ (m^2 - 4m + 13)e^{mx} &= 0 \\ m^2 - 4m + 13 &= 0. \end{aligned}$$

22 b)

$$\begin{aligned} m^2 - 4m + 13 &= 0 \\ (m-2)^2 - 4 + 13 &= 0 \\ (m-2)^2 &= -9 \\ m &= 2 \pm 3i. \end{aligned}$$

Thus the roots are in the form $a \pm bi$, where $a = 2, b = 3$.

23 a)

$$\begin{aligned}
y &= e^{(2+3i)x} \\
y' &= (2+3i)e^{(2+3i)x} \\
y'' &= (2+3i)^2 e^{(2+3i)x} = (-5+12i)e^{(2+3i)x} \\
y'' - 4y' + 13y &= (-5+12i)e^{(2+3i)x} - 4(2+3i)e^{(2+3i)x} + 13e^{(2+3i)x} \\
&= (-5+12i-8-12i+13)e^{(2+3i)x} \\
&= 0 = \text{RHS}.
\end{aligned}$$

23 b)

$$\begin{aligned}
y &= e^{(2-3i)x} \\
y' &= (2-3i)e^{(2-3i)x} \\
y'' &= (2-3i)^2 e^{(2-3i)x} = (-5-12i)e^{(2-3i)x} \\
y'' - 4y' + 13y &= (-5-12i)e^{(2-3i)x} - 4(2-3i)e^{(2-3i)x} + 13e^{(2-3i)x} \\
&= (-5-12i-8+12i+13)e^{(2-3i)x} \\
&= 0 = \text{RHS}.
\end{aligned}$$

23 c)

$$\begin{aligned}
y &= 5e^{(2+3i)x} + 2e^{(2-3i)x} \\
\frac{dy}{dx} &= 5(2+3i)e^{(2+3i)x} + 2(2-3i)e^{(2-3i)x} \\
\frac{d^2y}{dx^2} &= 5(2+3i)^2 e^{(2+3i)x} + 2(2-3i)^2 e^{(2-3i)x} \\
\text{LHS} &= \big(5(2+3i)^2 - 20(2+3i) + 65\big)e^{(2+3i)x} \\
&\quad + \big(2(2-3i)^2 - 8(2-3i) + 26\big)e^{(2-3i)x} \\
&= \big(5(4+12i-9) - (40+60i) + 65\big)e^{(2+3i)x} \\
&\quad + \big(2(4-12i-9) - (16-24i) + 26\big)e^{(2-3i)x} \\
&= (-25+60i-40-60i+65)e^{(2+3i)x} \\
&\quad + (-10-24i-16+24i+26)e^{(2-3i)x} \\
&= 0 = \text{RHS}.
\end{aligned}$$

24)

$$\begin{aligned}
e^{(2+3i)x} &= e^{2x}(\cos 3x + i\sin 3x) \\
e^{(2-3i)x} &= e^{2x}(\cos 3x - i\sin 3x) \\
Ae^{(2+3i)x} + Be^{(2-3i)x} &= e^{2x}\big((A+B)\cos 3x + (A-B)i\sin 3x\big).
\end{aligned}$$

Since the differential equation has real coefficients, for real-valued solutions choose $A = p + qi,\ B = p - qi$. Then $C = A + B = 2p$ and $D = (A - B)i = -2q$, which are real constants. Hence

$$Ae^{(2+3i)x} + Be^{(2-3i)x} = e^{2x}(C\cos 3x + D\sin 3x).$$

25 a)

$$\begin{aligned} y &= e^{2x}\cos 3x \\ y' &= e^{2x}(2\cos 3x - 3\sin 3x) \\ y'' &= e^{2x}(-5\cos 3x - 12\sin 3x) \\ y'' - 4y' + 13y &= e^{2x}[(-5 - 8 + 13)\cos 3x + (-12 + 12)\sin 3x] \\ &= 0 = \text{RHS}. \end{aligned}$$

25 b)

$$\begin{aligned} y &= e^{2x}\sin 3x \\ y' &= e^{2x}(2\sin 3x + 3\cos 3x) \\ y'' &= e^{2x}(-5\sin 3x + 12\cos 3x) \\ y'' - 4y' + 13y &= e^{2x}[(-5 - 8 + 13)\sin 3x + (12 - 12)\cos 3x] \\ &= 0 = \text{RHS}. \end{aligned}$$

25 c)

$$\begin{aligned} y &= 4e^{2x}\cos 3x \\ y' &= 4e^{2x}(2\cos 3x - 3\sin 3x) \\ y'' &= 4e^{2x}(-5\cos 3x - 12\sin 3x) \\ y'' - 4y' + 13y &= 4e^{2x}[(-5 - 8 + 13)\cos 3x + (-12 + 12)\sin 3x] \\ &= 0 = \text{RHS}. \end{aligned}$$

25 d)

$$\begin{aligned} y &= 5e^{2x}\sin 3x \\ y' &= 5e^{2x}(2\sin 3x + 3\cos 3x) \\ y'' &= 5e^{2x}(-5\sin 3x + 12\cos 3x) \\ y'' - 4y' + 13y &= 5e^{2x}[(-5 - 8 + 13)\sin 3x + (12 - 12)\cos 3x] \\ &= 0 = \text{RHS}. \end{aligned}$$

25 e)

$$
\begin{aligned}
y &= e^{2x}\cos 3x + e^{2x}\sin 3x \\
\frac{dy}{dx} &= -3e^{2x}\sin 3x + 2e^{2x}\cos 3x + 3e^{2x}\cos 3x + 2e^{2x}\sin 3x \\
&= 5e^{2x}\cos 3x - e^{2x}\sin 3x \\
\frac{d^2y}{dx^2} &= 10e^{2x}\cos 3x - 15e^{2x}\sin 3x - 2e^{2x}\sin 3x - 3e^{2x}\cos 3x \\
&= 7e^{2x}\cos 3x - 17e^{2x}\sin 3x \\
\text{LHS} &= (7e^{2x}\cos 3x - 17e^{2x}\sin 3x) - 4(5e^{2x}\cos 3x - e^{2x}\sin 3x) \\
&\quad + 13(e^{2x}\cos 3x + e^{2x}\sin 3x) \\
&= 7e^{2x}\cos 3x - 17e^{2x}\sin 3x - 20e^{2x}\cos 3x + 4e^{2x}\sin 3x \\
&\quad + 13e^{2x}\cos 3x + 13e^{2x}\sin 3x \\
&= (7 - 20 + 13)e^{2x}\cos 3x + (-17 + 4 + 13)e^{2x}\sin 3x \\
&= 0 = \text{RHS}.
\end{aligned}
$$

25 f)

$$
\begin{aligned}
y &= 4e^{2x}\cos 3x + 5e^{2x}\sin 3x \\
\frac{dy}{dx} &= -12e^{2x}\sin 3x + 8e^{2x}\cos 3x + 15e^{2x}\cos 3x + 10e^{2x}\sin 3x \\
&= 23e^{2x}\cos 3x - 2e^{2x}\sin 3x \\
\frac{d^2y}{dx^2} &= 46e^{2x}\cos 3x - 69e^{2x}\sin 3x - 4e^{2x}\sin 3x - 6e^{2x}\cos 3x \\
&= 40e^{2x}\cos 3x - 73e^{2x}\sin 3x \\
\text{LHS} &= (40e^{2x}\cos 3x - 73e^{2x}\sin 3x) - 4(23e^{2x}\cos 3x - 2e^{2x}\sin 3x) \\
&\quad + 13(4e^{2x}\cos 3x + 5e^{2x}\sin 3x) \\
&= 40e^{2x}\cos 3x - 73e^{2x}\sin 3x - 92e^{2x}\cos 3x + 8e^{2x}\sin 3x \\
&\quad + 52e^{2x}\cos 3x + 65e^{2x}\sin 3x \\
&= (40 - 92 + 52)e^{2x}\cos 3x + (-73 + 8 + 65)e^{2x}\sin 3x \\
&= 0 = \text{RHS}.
\end{aligned}
$$

26 a)

$$
\begin{aligned}
y &= e^{mx} \\
y' &= me^{mx} \\
y'' &= m^2e^{mx} \\
y'' - 4y' + 29y &= 0 \\
(m^2 - 4m + 29)e^{mx} &= 0 \\
m^2 - 4m + 29 &= 0.
\end{aligned}
$$

26 b)

$$\begin{aligned} m^2 - 4m + 29 &= 0 \\ (m-2)^2 - 4 + 29 &= 0 \\ (m-2)^2 &= -25 \\ m &= 2 \pm 5i. \end{aligned}$$

26 c)

$$\begin{aligned} y &= e^{2x}(A\cos 5x + B\sin 5x) \\ \frac{dy}{dx} &= e^{2x}(-5A\sin 5x + 5B\cos 5x) + 2e^{2x}(A\cos 5x + B\sin 5x) \\ &= e^{2x}((-5A+2B)\sin 5x + (2A+5B)\cos 5x) \\ \frac{d^2y}{dx^2} &= e^{2x}((-25A+10B)\cos 5x - (10A+25B)\sin 5x) \\ &\quad + 2e^{2x}((-5A+2B)\sin 5x + (2A+5B)\cos 5x) \\ &= e^{2x}((-25A+10B)\cos 5x - (10A+25B)\sin 5x) \\ &\quad + e^{2x}((-10A+4B)\sin 5x + (4A+10B)\cos 5x) \\ &= e^{2x}((-21A+20B)\cos 5x + (-20A-21B)\sin 5x). \end{aligned}$$

$$\begin{aligned} \text{LHS} &= e^{2x}((-21A+20B)\cos 5x + (-20A-21B)\sin 5x) \\ &\quad - 4e^{2x}((-5A+2B)\sin 5x + (2A+5B)\cos 5x) \\ &\quad + 29e^{2x}(A\cos 5x + B\sin 5x) \\ &= e^{2x}((-21A+20B)\cos 5x + (-20A-21B)\sin 5x) \\ &\quad + e^{2x}((20A-8B)\sin 5x + (-8A-20B)\cos 5x) \\ &\quad + e^{2x}(29A\cos 5x + 29B\sin 5x) \\ &= e^{2x}((-21A+20B-8A-20B+29A)\cos 5x \\ &\quad + (-20A-21B+20A-8B+29B)\sin 5x) \\ &= 0 = \text{RHS}. \end{aligned}$$

27 a)

$$\begin{aligned} y = e^{mx}, \quad y' = me^{mx}, \quad y'' &= m^2e^{mx} \\ y'' - 2y' + 5y &= 0 \\ (m^2 - 2m + 5)e^{mx} &= 0, \quad e^{mx} \neq 0 \\ m^2 - 2m + 5 &= 0 \\ (m-1)^2 - 1 + 5 &= 0 \\ (m-1)^2 &= -4 \\ m &= 1 \pm 2i \\ y &= e^x(A\cos 2x + B\sin 2x). \end{aligned}$$

27 b)

$$
\begin{aligned}
y = e^{mx}, \quad y' = me^{mx}, \quad y'' &= m^2e^{mx} \\
y'' + 6y' + 13y &= 0 \\
(m^2 + 6m + 13)e^{mx} &= 0, \quad e^{mx} \neq 0 \\
m^2 + 6m + 13 &= 0 \\
(m+3)^2 - 9 + 13 &= 0 \\
(m+3)^2 &= -4 \\
m &= -3 \pm 2i \\
y &= e^{-3x}(A\cos 2x + B\sin 2x).
\end{aligned}
$$

27 c)

$$
\begin{aligned}
y = e^{mx}, \quad y' = me^{mx}, \quad y'' &= m^2e^{mx} \\
y'' - 8y' + 25y &= 0 \\
(m^2 - 8m + 25)e^{mx} &= 0, \quad e^{mx} \neq 0 \\
m^2 - 8m + 25 &= 0 \\
(m-4)^2 - 16 + 25 &= 0 \\
(m-4)^2 &= -9 \\
m &= 4 \pm 3i \\
y &= e^{4x}(A\cos 3x + B\sin 3x).
\end{aligned}
$$

28 a)

$$
\begin{aligned}
y'' - 6y' + 73y &= 0 \\
m^2 - 6m + 73 &= 0 \\
(m-3)^2 - 9 + 73 &= 0 \\
(m-3)^2 &= -64 \\
m &= 3 \pm 8i \\
y &= e^{3x}(A\cos 8x + B\sin 8x).
\end{aligned}
$$

28 b)

$$
\begin{aligned}
16y'' - 80y' + 101y &= 0 \\
16m^2 - 80m + 101 &= 0 \\
16(m^2 - 5m) + 101 &= 0 \\
16\left((m - \frac{5}{2})^2 - \frac{25}{4}\right) + 101 &= 0 \\
16(m - \frac{5}{2})^2 - 100 + 101 &= 0
\end{aligned}
$$

$$
\begin{aligned}
(m-\frac{5}{2})^2 &= -\frac{1}{16} \\
m &= \frac{5}{2} \pm \frac{1}{4}i \\
y &= e^{5x/2}\left(A\cos\frac{x}{4} + B\sin\frac{x}{4}\right).
\end{aligned}
$$

29)

$$
\begin{aligned}
y'' - 6y' + 10y &= 0 \\
m^2 - 6m + 10 &= 0 \\
(m-3)^2 - 9 + 10 &= 0 \\
(m-3)^2 &= -1 \\
m &= 3 \pm i \\
y &= e^{3x}(A\cos x + B\sin x).
\end{aligned}
$$

Substitute $y = 4, x = 0$:

$$
\begin{aligned}
4 &= e^0(A\cos 0 + B\sin 0) \\
4 &= A + 0 \\
A &= 4.
\end{aligned}
$$

Substitute $y = 2, x = \pi/2$:

$$
\begin{aligned}
2 &= e^{3\pi/2}(4\cos\frac{\pi}{2} + B\sin\frac{\pi}{2}) \\
2 &= e^{3\pi/2}B \\
B &= 2e^{-3\pi/2}.
\end{aligned}
$$

Particular solution: $y = e^{3x}(4\cos x + 2e^{-3\pi/2}\sin x)$.

30)

$$
\begin{aligned}
\ddot{x} - 4\dot{x} + 13x &= 0 \\
m^2 - 4m + 13 &= 0 \\
(m-2)^2 - 4 + 13 &= 0 \\
(m-2)^2 &= -9 \\
m &= 2 \pm 3i \\
x &= e^{2t}(A\cos 3t + B\sin 3t).
\end{aligned}
$$

Substitute $x = 2, t = 0$:

$$
\begin{aligned}
2 &= e^0(A\cos 0 + B\sin 0) \\
A &= 2.
\end{aligned}
$$

Then $x = e^{2t}(2\cos 3t + B\sin 3t)$, so

$$
\dot{x} = e^{2t}(-6\sin 3t + 3B\cos 3t) + 2e^{2t}(2\cos 3t + B\sin 3t)
$$

$$= e^{2t}((-6+2B)\sin 3t + (4+3B)\cos 3t).$$

Substitute $\dot{x} = 1, t = 0$:

$$\begin{aligned} 1 &= e^0((-6+2B)\sin 0 + (4+3B)\cos 0) \\ 1 &= 4+3B \\ B &= -1. \end{aligned}$$

Particular solution: $x = e^{2t}(2\cos 3t - \sin 3t)$.

31 a)

$$\begin{aligned} y'' - 10y' + 16y &= 0 \\ m^2 - 10m + 16 &= 0 \\ (m-2)(m-8) &= 0 \\ m = 2 \text{ or } m &= 8 \\ y &= Ae^{2x} + Be^{8x}. \end{aligned}$$

31 b)

$$\begin{aligned} y'' - 18y' + 97y &= 0 \\ m^2 - 18m + 97 &= 0 \\ (m-9)^2 - 81 + 97 &= 0 \\ (m-9)^2 &= -16 \\ m &= 9 \pm 4i \\ y &= e^{9x}(A\cos 4x + B\sin 4x). \end{aligned}$$

31 c)

$$\begin{aligned} \ddot{x} - 11\dot{x} - 60x &= 0 \\ m^2 - 11m - 60 &= 0 \\ (m+4)(m-15) &= 0 \\ m = -4 \text{ or } m &= 15 \\ x &= Ae^{-4t} + Be^{15t}. \end{aligned}$$

31 d)

$$\begin{aligned} 121y'' - 110y' + 25y &= 0 \\ 121m^2 - 110m + 25 &= 0 \\ (11m-5)^2 & \\ m &= \frac{5}{11} \\ y &= Axe^{5x/11} + Be^{5x/11}. \end{aligned}$$

31 e)

$$
\begin{aligned}
9\ddot{x} + 42\dot{x} + 49x &= 0 \\
9m^2 + 42m + 49 &= 0 \\
(3m+7)^2 &= 0 \\
m &= -\frac{7}{3} \\
x &= Ate^{-7t/3} + Be^{-7t/3}.
\end{aligned}
$$

31 f)

$$
\begin{aligned}
y'' - 30y' + 225y &= 0 \\
m^2 - 30m + 225 &= 0 \\
(m-15)^2 & \\
m &= 15 \\
y &= Axe^{15x} + Be^{15x}.
\end{aligned}
$$

31 g)

$$
\begin{aligned}
y'' + y &= 0 \\
m^2 + 1 &= 0 \\
m^2 &= -1 \\
m &= \pm i \\
y &= A\cos x + B\sin x.
\end{aligned}
$$

31 h)

$$
\begin{aligned}
2\ddot{x} + 2\dot{x} + 5x &= 0 \\
2m^2 + 2m + 5 &= 0 \\
2(m^2 + m) + 5 &= 0 \\
2((m+\frac{1}{2})^2 - \frac{1}{4}) + 5 &= 0 \\
2(m+\frac{1}{2})^2 - \frac{1}{2} + 5 &= 0 \\
(m+\frac{1}{2})^2 &= -\frac{9}{4} \\
m &= -\frac{1}{2} \pm \frac{3}{2}i \\
x &= e^{-t/2}(A\cos\frac{3}{2}t + B\sin\frac{3}{2}t).
\end{aligned}
$$

31 i)

$$\begin{aligned} y'' + 6y' + 109y &= 0 \\ m^2 + 6m + 109 &= 0 \\ (m+3)^2 - 9 + 109 &= 0 \\ (m+3)^2 &= -100 \\ m &= -3 \pm 10i \\ y &= e^{-3x}(A\cos 10x + B\sin 10x). \end{aligned}$$

31 j)

$$\begin{aligned} 81y'' + y &= 0 \\ 81m^2 + 1 &= 0 \\ m^2 &= -\frac{1}{81} \\ m &= \pm\frac{1}{9}i \\ y &= A\cos\frac{x}{9} + B\sin\frac{x}{9}. \end{aligned}$$

31 k)

$$\begin{aligned} y'' + 5y' - 36y &= 0 \\ m^2 + 5m - 36 &= 0 \\ (m+9)(m-4) & \\ m &= -9 \text{ or } m = 4 \\ y &= Ae^{-9x} + Be^{4x}. \end{aligned}$$

31 l)

$$\begin{aligned} y'' + 16y' + 64y &= 0 \\ m^2 + 16m + 64 &= 0 \\ (m+8)^2 &= 0 \\ m &= -8 \\ y &= Axe^{-8x} + Be^{-8x}. \end{aligned}$$

31 m)

$$\begin{aligned} 49\ddot{x} + 25x &= 0 \\ 49m^2 + 25 &= 0 \end{aligned}$$

$$m^2 = -\frac{25}{49}$$
$$m = \pm\frac{5}{7}i$$
$$x = A\cos\frac{5}{7}t + B\sin\frac{5}{7}t.$$

31 n)

$$30y'' + 49y' + 20y = 0$$
$$30m^2 + 49m + 20 = 0$$
$$(5m+4)(6m+5) = 0$$
$$m = -\frac{4}{5} \text{ or } m = -\frac{5}{6}$$
$$y = Ae^{-4x/5} + Be^{-5x/6}.$$

31 o)

$$y'' + 400y = 0$$
$$m^2 + 400 = 0$$
$$m^2 = -400$$
$$m = \pm 20i$$
$$y = A\cos 20x + B\sin 20x.$$

31 p)

$$\ddot{x} + 32x = 0$$
$$m^2 + 32 = 0$$
$$m^2 = -32$$
$$m = \pm 4\sqrt{2}i$$
$$x = A\cos(4\sqrt{2}t) + B\sin(4\sqrt{2}t).$$

31 q)

$$32y'' - 12y' - 35y = 0$$
$$32m^2 - 12m - 35 = 0$$
$$(8m+7)(4m-5)$$
$$m = -\frac{7}{8} \text{ or } m = \frac{5}{4}$$
$$y = Ae^{-7x/8} + Be^{5x/4}.$$

32 a)

$$\begin{aligned} y &= 5e^{-3x} + 6e^{4x} \\ m &= -3 \text{ or } m = 4 \\ \text{Auxiliary equation: } (m+3)(m-4) &= 0 \\ m^2 - m - 12 &= 0 \\ \text{Differential equation: } y'' - y' - 12y &= 0. \end{aligned}$$

32 b)

$$\begin{aligned} y &= e^{6x}(2\cos 5x + 9\sin 5x) \\ m &= 6 \pm 5i \\ \text{Auxiliary equation: } (m-6)^2 &= -25 \\ m^2 - 12m + 61 &= 0 \\ \text{Differential equation: } y'' - 12y' + 61y &= 0. \end{aligned}$$

32 c)

$$\begin{aligned} y &= 6xe^{14x} - 5e^{14x} \\ m &= 14 \\ \text{Auxiliary equation: } (m-14)^2 &= 0 \\ m^2 - 28m + 196 &= 0 \\ \text{Differential equation: } y'' - 28y' + 196y &= 0. \end{aligned}$$

32 d)

$$\begin{aligned} y &= 15\cos^2\frac{2}{3}x + 6\sin^2\frac{2}{3}x \\ m &= \pm\frac{2}{3}i \\ \text{Auxiliary equation: } m^2 &= -\frac{4}{9} \\ m^2 + \frac{4}{9} &= 0 \\ \text{Differential equation: } y'' + \frac{4}{9}y &= 0. \end{aligned}$$

33)

$$\begin{aligned} y'' - y' - 6y &= 0 \\ m^2 - m - 6 &= 0 \end{aligned}$$

$$(m+2)(m-3) = 0$$
$$m = -2 \text{ or } m = 3$$
$$y = Ae^{-2x} + Be^{3x}.$$

Substitute $y = 10, x = 0$:

$$10 = Ae^0 + Be^0$$
$$A + B = 10. \tag{1}$$

Substitute $y = 49, x = \ln 2$:

$$49 = Ae^{-2\ln 2} + Be^{3\ln 2}$$
$$49 = \frac{1}{4}A + 8B. \tag{2}$$

Solving (1) and (2), $A = 4, B = 6$. Particular solution $y = 4e^{-2x} + 6e^{3x}$.

34)

$$y'' - 2y' + 2y = 0$$
$$m^2 - 2m + 2 = 0$$
$$(m-1)^2 - 1 + 2 = 0$$
$$(m-1)^2 = -1$$
$$m = 1 \pm i$$
$$y = e^x(A\cos x + B\sin x).$$

Substitute $y = 2, x = 0$:

$$2 = e^0(A\cos 0 + B\sin 0)$$
$$A = 2.$$

So $y = e^x(2\cos x + B\sin x)$, giving

$$\frac{dy}{dx} = e^x(-2\sin x + B\cos x) + e^x(2\cos x + B\sin x)$$
$$= e^x((B+2)\cos x + (-2+B)\sin x).$$

Stationary point when $dy/dx = 0, x = \pi/6$:

$$0 = e^{\pi/6}((B+2)\cos\frac{\pi}{6} + (-2+B)\sin\frac{\pi}{6})$$
$$0 = (B+2)\cos\frac{\pi}{6} + (-2+B)\sin\frac{\pi}{6}$$
$$0 = \frac{\sqrt{3}}{2}(B+2) + \frac{1}{2}(-2+B)$$
$$0 = \sqrt{3}(B+2) + (-2+B)$$
$$0 = \sqrt{3}B + 2\sqrt{3} - 2 + B$$
$$2 - 2\sqrt{3} = (\sqrt{3}+1)B$$
$$B = \frac{2-2\sqrt{3}}{\sqrt{3}+1} = -4 + 2\sqrt{3}.$$

Particular solution:

$$y = e^x(2\cos x + (-4+2\sqrt{3})\sin x).$$